FlowVR — Inducing and Recovering Flow State in Virtual Reality

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***Abstract*—Flow State is a highly focused cognitive condition associated with enhanced performance, learning, and intrinsic motivation. This paper presents FlowVR, a virtual reality research system designed to examine whether flow can be reliably induced through immersive motor interaction and how users recover following an intentional disruption of attention. Participants engaged in a repetitive, embodied task designed to sustain continuous engagement while supporting measurable interruption and recovery. Behavioral performance, subjective experience, and electroencephalography (EEG) data were collected during controlled study sessions. Results suggest that immersive VR interaction can support sustained engagement consistent with Flow State and that recovery following disruption exhibits meaningful individual variation. While EEG data quality was impacted by hardware instability, subjective and behavioral findings provide insight into recovery dynamics that are underexplored in existing flow research. These findings contribute to understanding how immersive systems can support both flow induction and recovery.**

Index Terms—flow state, virtual reality, immersion, attention recovery

I. Introduction

Virtual reality (VR) systems are increasingly used in education, training, and productivity-oriented applications. However, sustaining user focus over time remains challenging, particularly in the presence of interruptions. Poorly designed interactions, cognitive overload, and external disruptions can significantly reduce immersion and performance.

Flow State, first introduced by Csikszentmihalyi [1], describes a mental condition characterized by deep focus, intrinsic motivation, and a balance between challenge and skill. While flow has been studied extensively in psychology and human–computer interaction, comparatively little work has examined how flow may be disrupted and subsequently recovered, particularly in immersive environments.

VR presents a unique opportunity to study these dynamics due to its capacity for embodied interaction, sensory immersion, and experimental control. Despite this potential, recovery dynamics following flow disruption remain underexplored.

This work addresses that gap by investigating both flow induction and recovery within a controlled VR task that allows intentional disruption.

A. Contributions

This paper makes the following contributions:

- A VR-based experimental framework for studying flow induction and recovery.
- A novel disruption mechanism that enables controlled interruption of immersive flow.
- Empirical observations of post-disruption recovery behavior using behavioral, subjective, and EEG measures.
- Design insights for embodied VR tasks that align with Flow Theory.

Fig. 1. FlowVR tennis court environment showing user position, ball basket, and interaction space.

II. Literature Review

The literature selected for this review encompasses research on both flow state and electroencephalography (EEG), enabling an integrated perspective on experiential and neural approaches to flow. Together, these works provide a foundation for understanding how flow has been conceptualized and measured in prior research, while also informing flow-related neural activity

Flow Theory originates with Csikszentmihalyi [1], who describes flow as a balance between challenge and skill that produces deep focus and intrinsic motivation. This foundation has influenced much of the modern work surrounding engagement in digital environments. Studies in games and interactive media have built on this, showing how responsive controls, clear feedback, and stable interaction loops support immersion and help users reach a deeper level of focus [3], [4], [11]. These works emphasize that flow is not only about attention, but that even minute inconsistencies can quickly break immersion. This is a motivator for many of the design choices in FlowVR, whether it is physics behavior, audio cues, or haptic feedback; all were tuned to push for as much immersion as possible.

Looking past interaction design, research has also gone into how flow can be intentionally induced or disrupted. Zielke and Berger [2] show noticeable changes in behavior and subjective experience when flow is broken, and that recovery can vary significantly across individuals. Work in educational and gamified contexts further shows that clear goals and immediate feedback can make flow easier to reach and maintain [6]. This is why we use a controlled disruption mechanic to examine how users recover flow after interruption.

To link conceptual understanding with empirical investigation, we also examined EEG studies, as they offer insights into a users' cognitive states.Several studies show that EEG patterns usually have shifts in alpha, theta, or gamma activity during immersive states [5], [13]–[15]. These findings suggest that consistent interaction creates measurable changes in attention and cognitive networks. Hang et al. [15] and Metin et al. [14] pinpoint that frontal theta activity is important during focused engagement, which lines up with our interest in them. Although our EEG signal quality was limited, the literature supports the idea that neural changes related to flow can be captured even with lightweight or single-channel setups. The influence of these were inconsequential as compared to Cortes, C(2023) "An EEG-based experiment on VR sickness and postural instability while walking in virtual environments" [8]. This study, which focused on VR and EEG, demonstrated the feasibility of measuring user engagement in immersive environments, providing a foundation that ensured measuring flow in virtual environments was possible.

Domain-specific VR research shows that immersive tasks are able to consistently induce flow. Work in robotic VR operations and emotional interactive VR experiences demonstrate that when there are clear goals and strong sensory feedback, users tend to report high engagement and show behavior consistent with flow [9], [12]. There is a big reliance on stable sensory and cognitive coupling, which is tied to how well user actions are gathered. Silva [7] explains that consistent and uninterrupted interaction is important for stable neural patterns, since noise and breaks in feedback can cause interference with engagement. This aligns with what we gathered in FlowVR where users quickly fell into a stable rhythm during the experience, and the disruption immediately broke that.

III. Methods and System Design

A. System Architecture

FlowVR consists of three integrated subsystems: (1) an interaction subsystem capturing controller and headset motion, (2) a physics and feedback subsystem governing object behavior and haptics, and (3) a neural data subsystem using an OpenBCI EEG cap. The system was developed in Unity 6000.2.6f1 using the OpenXR XR Interaction Toolkit and Unity's High Definition Rendering Pipeline, and deployed on Meta Quest 3 hardware.

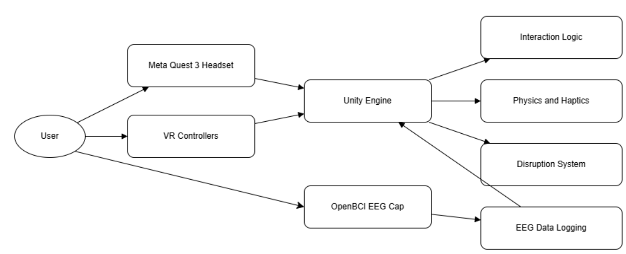

Fig. 2. System architecture of FlowVR showing VR interaction, physics simulation, and EEG data acquisition and analysis subsystems.

B. Task Design and Rationale

The tennis ball bouncing task was selected to align closely with Flow Theory. The task exhibits:

- **Low learning curve**, allowing rapid engagement.

- **Continuous motor involvement**, sustaining attention.
- **Natural VR embodiment**, mapping real-world motion to virtual action.
- **Scalable difficulty**, enabling self-regulated challenge.

These properties make the task well-suited for inducing and maintaining flow.

C. XR Interaction Design

Our task design focused on creating natural, immersive player interactions. While we initially used the XR Origins prefab to control VR, it did not fully align with our design goals. To better support flow, we implemented several modifications aimed at improving interaction fidelity and realism, giving players a more intuitive, experience. These modifications came mostly in the form of restrictions. By default, the XR Toolkit allows several different ways for users to interact with objects in VR, but too many interaction options would not be immersive. For our use case, it was important that users could only walk up to objects and grab them. So we used the near-far interactor and modified it to only support near interactions. As a second restriction, we disabled thumb stick movement and turning, since allowing these actions would have exponentially increased the likelihood of cybersickness and disrupted immersion, potentially harming our results. Finally, we replaced the default controller models with neutral designs to avoid distracting looking-grab behavior. These XR modifications established a stable foundation for controlling the simulator and were a key building block for designing our physics systems.

D. Physics Simulation Design

To support the induction and maintenance of a flow state during VR interaction, bouncing behavior was implemented using Unity's physics materials to ensure consistent, predictable, and perceptually coherent object responses. Bouncing behavior parameters were manually tuned to achieve a level of difficulty that supported users' ability to enter a flow state while maintaining perceptual realism. Contributing to this realism, additional racket tweaks were used. Velocity tracking was added to allow users to control the ball's force through their own movement speed, letting them modulate challenge and maintain a sense of agency during the task. Multiple box colliders were used to approximate the racket shape, providing more predictable collision behavior than sphere or mesh colliders, which helped maintain immersion and prevented cognitive distractions.

Fig. 3. First-person view of racket–ball interaction illustrating embodied control and feedback.

E. Haptic Feedback Design

Haptic feedback was implemented to add more immersion during the interaction and support consistent engagement. Controller vibration was triggered only when the racket made contact with the ball and was actively held by the user, so that haptic cues were consistently intentional rather than incidental collisions or other unwanted results. A minimum collision velocity threshold was set so that light collisions did not activate a vibration; however, this is generally unnoticed and has no significant impact. Haptic amplitude was scaled based on collision velocity and set within a range to maintain consistency across repeated interactions.

To avoid overstimulation during sequences of hits, a short global cooldown was added between haptics as a precaution. During the disruption, collision detection is disabled, which results in the complete removal of haptic feedback. Being disabled during the disruption adds to the effectiveness of breaking the Flow Slate.

F. Audio Feedback Design

Audio feedback complemented haptics through collisions as well. Distinct audio clips were used for racket impacts and ball bounces, with playback volume scaled by collision velocity for the sake of immersion. Similarly to the haptics, a minimum velocity threshold cuts out accidental or weak collisions, preventing excessive and unwanted sounds. This helps maintain clean and immersive audio during the experience.

To reduce noticeable repetitiveness and add to the immersion, a subtle pitch randomization was applied independently to each sound event. A shared global cooldown limited the frequency of audio playback across all balls, preventing clutter if collisions ever happened rapidly. Together with haptic feedback, audio cues build up a sense of rhythm and support sustained attention and immersion. During the disruption, the removal of impact sounds further adds to the flow state break along with the haptics.

G. Disruption System

To investigate the dynamics of flow in VR, we implemented a novel disruption mechanism designed to intentionally interrupt the user's flow state. 120 seconds after the first successful bounce interaction, ball–racket collisions were disabled for 30 seconds. This manipulation breaks sensorimotor coupling without altering the environment visually, enabling precise observation of recovery dynamics. Recovery after disruption remains underexplored in flow research and represents a key novelty of this work.

H. Data Collection and Analysis

The study was conducted at the University of Oklahoma with a total of seven participants (N = 7), consisting of both undergraduate and graduate students recruited from computer science and related disciplines. Most participants had prior experience with basic virtual reality interaction, though none had prior exposure to the FlowVR system. Each participant completed a single experimental session under controlled laboratory conditions.

Data collection incorporated three complementary sources. First, task performance metrics were recorded directly within the VR environment, including duration of sustained interaction, number of successful ball bounces, disruption onset time, and recovery time following disruption. Recovery time was defined as the duration required for participants to re-establish consistent, stable interaction after the disruption period ended.

Second, neural data were collected using a 16-node OpenBCI electroencephalography (EEG) headset. EEG data were time-synchronized with in-game events, including flow onset and disruption onset, to enable exploratory examination of neural activity associated with sustained engagement and recovery.

Third, participants completed pre- and post-experience questionnaires designed to capture subjective aspects of immersion, perceived engagement, stress, distraction, comfort, and recovery difficulty. Questionnaire items were measured using Likert-scale responses. All questionnaire responses were converted to numerical values prior to analysis.

Given the exploratory and pilot nature of the study, data analysis focused on descriptive statistics and relational trends rather than inferential hypothesis testing. Pearson correlation coefficients were computed to examine relationships between subjective survey measures and behavioral performance metrics, particularly recovery time. Correlation results are reported to identify strong positive and negative relationships and are interpreted as indicative trends rather than causal or generalizable findings.

I. EEG Integration

Electroencephalogram (EEG) is what we used in order to monitor theta waves during the simulation. More specifically we used Cyton + Daisy Bio-sensing Boards with a Gelfree BCI Electrode Cap. We wanted to monitor these theta waves in order to see a clear break in flow after the disruption. In order to get a constant stream of data we designed a custom clip in order to have the headset's board attach to the users pants allowing for movement. We needed to use both the Quest 3 and the EEG at the same time which came into a conflict on the VR headset rubbing on some of the sixteen nodes that we used for data collection. To remedy this we took off the top strap and held the VR headset up by its side straps. This allowed for all the nodes to be exposed and not in contact with the headset to reduce noise made by the user's movement. Being able to see live user data was a great help in the testing phase because it allowed us in real time to see how the participant was doing. This stream of data was also important because it allowed us to make sure the headset was working at all time, and made sure all nodes where firing correctly.

IV. Results and Evaluation

Across all participants, sustained interaction within the FlowVR environment was associated with high reported levels of immersion and engagement. Participants consistently described the tennis ball bouncing task as intuitive and absorbing, enabling continuous attention without requiring explicit instruction or cognitive effort once interaction was established.

The scripted disruption was perceived by all participants and resulted in an immediate breakdown of interaction consistency. Following disruption, recovery time varied notably across individuals, ranging from rapid re-engagement to prolonged difficulty re-establishing stable interaction. This variability suggests individual differences in attentional resilience and recovery capacity.

Analysis of questionnaire responses revealed multiple strong relationships among flow-related subjective measures. Pearson correlation analysis identified several strong positive correlations ($r \geq 0.75$) between items assessing immersion, engagement, task fluency, and perceived control. These findings suggest internal consistency among measures intended to capture Flow State and support the construct validity of the survey instrument.

Strong negative correlations ($r \leq -0.75$) were also observed between flow-related measures and items assessing distraction, discomfort, and perceived interruption. Participants who reported higher levels of distraction or discomfort tended to report lower immersion and longer recovery times. These inverse relationships are consistent with theoretical expectations derived from Flow Theory.

In addition to this skill was not factored enough into development of the simulator. Some users struggled with performing the task, either due to discomfort with VR systems or poor coordination. At least two participants reported having extreme difficulty using the simulator; one was excluded from the data due to cybersickness before any meaningful interaction had begun; this users cybersickness persisted for 4 hours following the test. Several participants even some with VR experience struggled to keep a hold on the racket dropping it multiple times creating a self induced disruption.

Behavioral performance metrics further complemented subjective findings. Participants reporting higher immersion and engagement generally demonstrated faster recovery following disruption, whereas those reporting increased stress or distraction required longer periods to regain consistent interaction. Due to the limited sample size, these results are presented descriptively and are not intended to support inferential statistical claims.

V. Discussion

The findings of this study suggest that immersive, embodied VR tasks can effectively support sustained engagement consistent with flow state. The tennis ball bouncing task successfully encouraged continuous motor involvement, minimized cognitive overhead, and allowed participants to self-regulate challenge through interaction precision. These properties align closely with established conditions for flow, including clear goals, immediate feedback, and a balance between skill and challenge.

A central contribution of this work is the explicit examination of recovery following flow disruption. While prior research has focused primarily on inducing flow or maximizing immersion, recovery dynamics remain underexplored. The disruption system implemented in FlowVR enabled controlled interruption of sensorimotor coupling without altering the visual environment, providing a novel mechanism for observing recovery behavior. The observed variability in recovery time highlights recovery as a meaningful dimension of flow that extends beyond initial induction.

EEG data collection introduced additional methodological complexity. Hardware instability associated with the OpenBCI system limited the reliability of neural signals and constrained interpretation of EEG-derived measures. Explicit acknowledgment of this limitation is consistent with best scientific practice and underscores the challenges of integrating mobile EEG systems with immersive VR. As a result, EEG data were treated as exploratory, and primary conclusions were drawn from behavioral and subjective measures.

Despite these limitations, the convergence of self-reported experience and performance metrics provides preliminary evidence supporting the feasibility of FlowVR as a research platform. Future work should prioritize more stable neural acquisition systems, larger participant samples, and longitudinal designs to further investigate flow recovery dynamics.

VI. Conclusion

This paper presented FlowVR, a virtual reality research framework designed to investigate both the induction and recovery of Flow State through immersive, embodied interaction. By integrating a repetitive motor task, a novel disruption mechanism, and multimodal data collection, the study provides insight into how flow-like engagement can be sustained and how individuals recover following interruption.

Results from a pilot study involving seven participants indicate that immersive VR interaction can support strong subjective immersion and engagement, while recovery following disruption exhibits meaningful individual variation. Although EEG data reliability was limited, behavioral and survey-based findings highlight recovery dynamics as a promising and underexplored dimension of Flow Theory.

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